Electrical Workshop II

INDUCTION MOTOR WINDING

Contents

- ***Introduction to Induction motor;***
- ***Three phase induction motor winding types and arrangement;***
- ***Design of three phase induction motor winding types;***
- ***Winding of coils using manual winding machine;***
- ***Assembly of the winding to the stator of the motor and finally performing all necessary tests that are done by maintenance personnel***

Introduction to Induction Motor

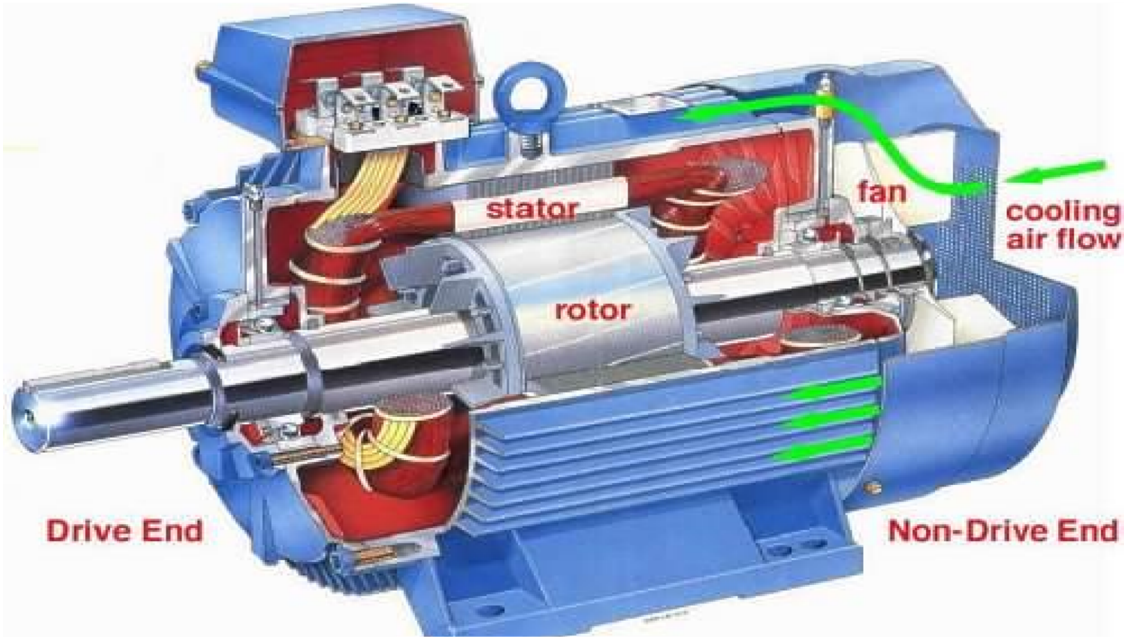
- Three-phase induction motors are the most common and frequently encountered machines in industry;
  - Simple design, low-price, easy maintenance
  - Wide range of power ratings: fractional horsepower to 10 MW
  - Run essentially as constant speed from no-load to full load
  - Its speed depends on the frequency of the power source
  - Requires a variable-frequency power-electronic drive for optimal speed control

Construction of IM

- An induction motor has two main parts
  - **A stationary stator**
    - Consisting of a steel frame that supports a hollow, cylindrical core
    - Core, constructed from stacked laminations, having a number of **evenly spaced slots**, providing the space for the stator winding
  - **A revolving rotor**
    - Composed of punched laminations, stacked to create a series of rotor slots, providing space for the rotor winding
    - One of two types of rotor windings
    - Conventional 3-phase windings made of insulated wire (wound-rotor) » similar to the winding on the stator
    - Aluminum bus bars shorted together at the ends by two aluminum rings, forming a squirrel-cage shaped circuit (squirrel-cage)

➢Two basic design types depending on the rotor design

- **Squirrel-cage:** conducting bars laid into slots and ***shorted at both ends*** by shorting rings.
- **Wound-rotor:** complete set of three-phase windings exactly as the stator. Usually Y-connected, the ends of the three rotor wires are connected to 3 slip rings on the rotor shaft. In this way, the rotor circuit is accessible.





Squirrel cage rotor

Wound rotor

Notice the slip rings



Principle of Operation

- A Rotating Magnetic field (**RMF**) is set up in the stator when a 3- Phase supply is given.
- This rotating magnetic field cuts the rotor windings and produces an induced voltage in the rotor windings.
- Due to the fact that the rotor windings are short circuited, for both squirrel cage and wound-rotor, and induced current flows in the rotor windings.
- The rotor current produces another magnetic field.
- A torque is produced as a result of the interaction of those two magnetic fields.

$$\tau_{ind} = kB_R \times B_s$$

Where τ_{ind} is the induced torque and B_R and B_S are the magnetic flux densities of the rotor and the stator respectively

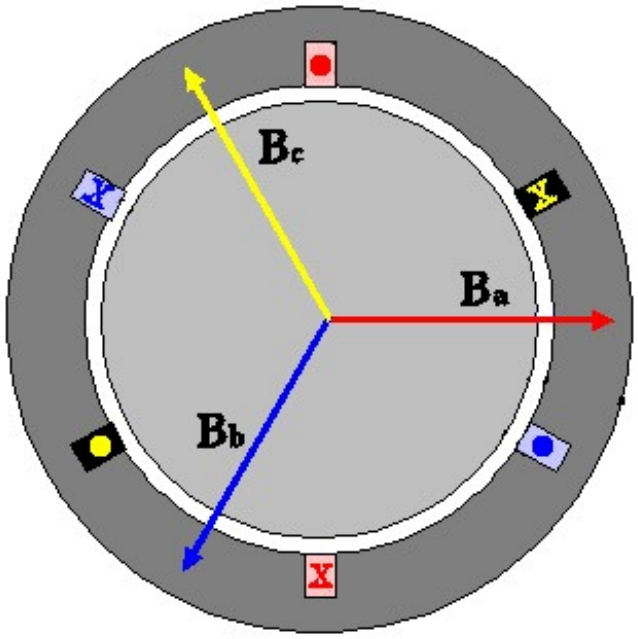
Rotating Magnetic Field

- Balanced three phase windings, i.e. mechanically displaced 120 degrees form each other, fed by balanced three phase source
- A rotating magnetic field with constant magnitude is produced, rotating with a speed

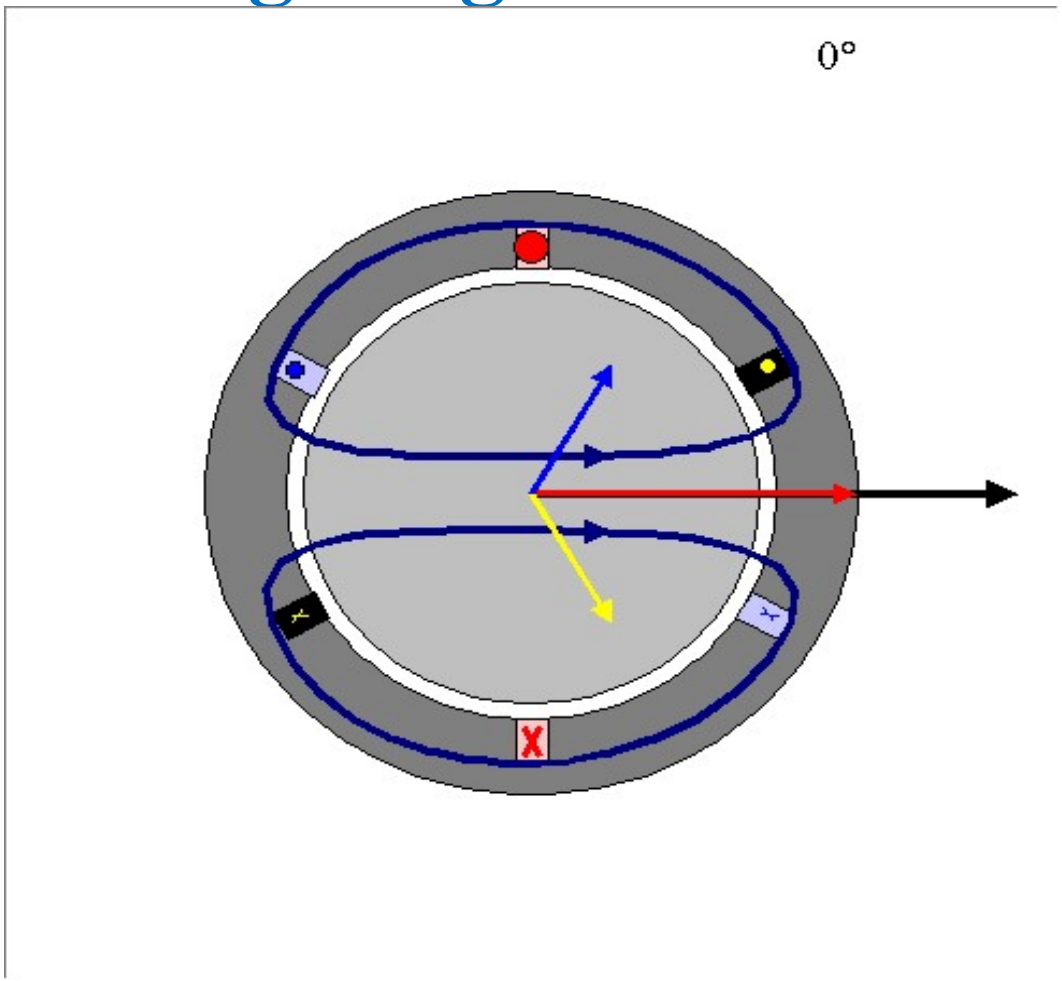
$$n_{sync} = \frac{120 f_e}{P} \quad rpm$$

Where

- f_e is the supply frequency and
- P is the no. of poles and
- n_{sync} is called the synchronous speed in *rpm* (revolutions per minute)



Rotating Magnetic Field



Induction motor speed

- At what speed will the IM run?
  - Can the IM run at the synchronous speed, why?
  - If rotor runs at the synchronous speed, which is the same speed of the rotating magnetic field, then the rotor will appear stationary to the rotating magnetic field and the rotating magnetic field will not cut the rotor.
  - So, no induced current will flow in the rotor and no rotor magnetic flux will be produced so no torque is generated and the rotor speed will fall below the synchronous speed
  - When the speed falls, the rotating magnetic field will cut the rotor windings and a torque is produced

Induction motor speed

- So, the IM will always run at a speed **lower** than the synchronous speed.
- The difference between the motor speed and the synchronous speed is called the ***Slip***

$$n_{slip} = n_{sync} - n_m$$

Where n_{slip}= slip speed

n_{sync}= speed of the magnetic field

n_m = mechanical shaft speed of the motor

The Slip

$$s = \frac{n_{sync} - n_m}{n_{sync}}$$

Where, ***s*** is the *slip*

Notice that : **if the rotor runs at synchronous speed**

$$s = 0$$

if the rotor is stationary

$$s = 1$$

❖ Slip may be expressed as a percentage by multiplying the above eq. by 100, notice that the slip is a ratio and doesn't have units

Frequency

- The frequency of the voltage induced in the ***rotor*** is given by

$$f_r = \frac{P \times nslip}{120}$$

Where f_r = the rotor frequency (Hz)

P = number of stator poles

$nslip$ = slip speed (rpm)

$$f_r = \frac{P \times (n_s - n_m)}{120}$$

$$= \frac{P \times s n_s}{120} = s f_e$$

- What would be the ***frequency of the rotor's*** induced voltage at any speed n_m?

$$f_r = s\, f_e$$

- When the **rotor is blocked** (s=1) , the frequency of the induced voltage is equal to the supply frequency.
- On the other hand, if the rotor runs at synchronous speed (s = 0), the frequency will be zero.

Torque

- While the input to the induction motor is electrical power, its output is mechanical power and for that we should know some terms and quantities related to mechanical power.
- Any mechanical load applied to the motor shaft will introduce a ***Torque*** on the motor shaft.
- This torque is related to the motor output power and the rotor speed

$$\tau_{load} = \frac{P_{out}}{\omega_m} \quad N.m$$

and

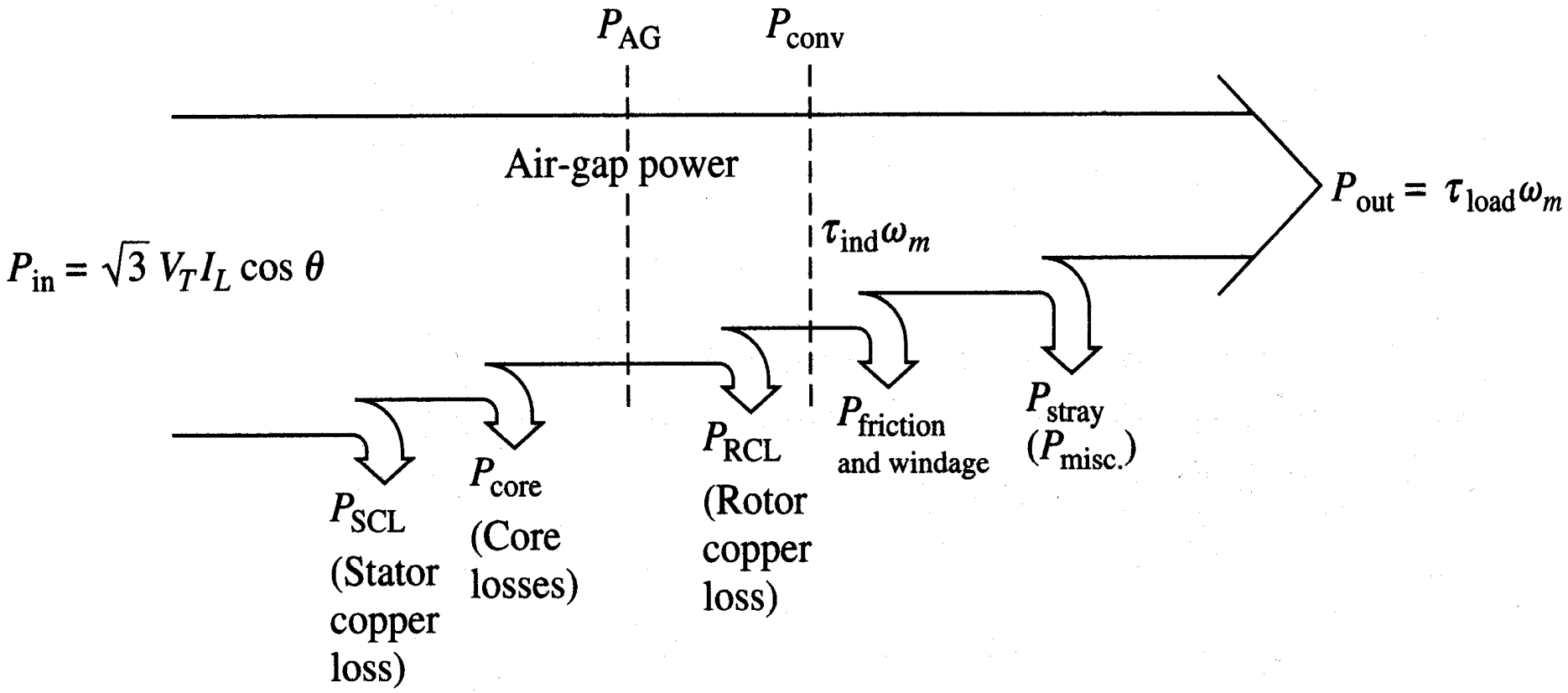
$$\omega_m = \frac{2\pi n_m}{60} \quad rad/s$$

Horse Power

- Another unit used to measure mechanical power is the **horse power.**
- It is used to refer to the mechanical output power of the motor.
- Since we, as an electrical engineers, deal with **watts** as a unit to measure electrical power, there is a relation between horse power and watts

$$hp = 746\ watts$$

Power flow in induction motor



Power Relations

$$P_{in} = \sqrt{3}\, V_L I_L \cos\theta = 3\, V_{ph} I_{ph} \cos\theta$$

$$P_{SCL} = 3\ I_1^2 R_1$$

$$P_{AG} = P_{in} - (P_{SCL} + P_{core})$$

$$P_{RCL} = 3 I_2^2 R_2$$

$$P_{conv} = P_{AG} - P_{RCL}$$

$$\tau_{ind} = \frac{P_{conv}}{\omega_m}$$

$$P_{out} = P_{conv} - (P_{f+w} + P_{stray})$$

Components of 3-phase Induction Motor

1. **Frame**
2. **Stator**
3. **Stator Winding**
4. **Rotor**
5. **Rotor Winding**
6. **Cooling Fan**
7. **Bearings Frame:**

➢ **Frame:** provides mechanical support to the stator and rotor.

- It is made from casting materials.

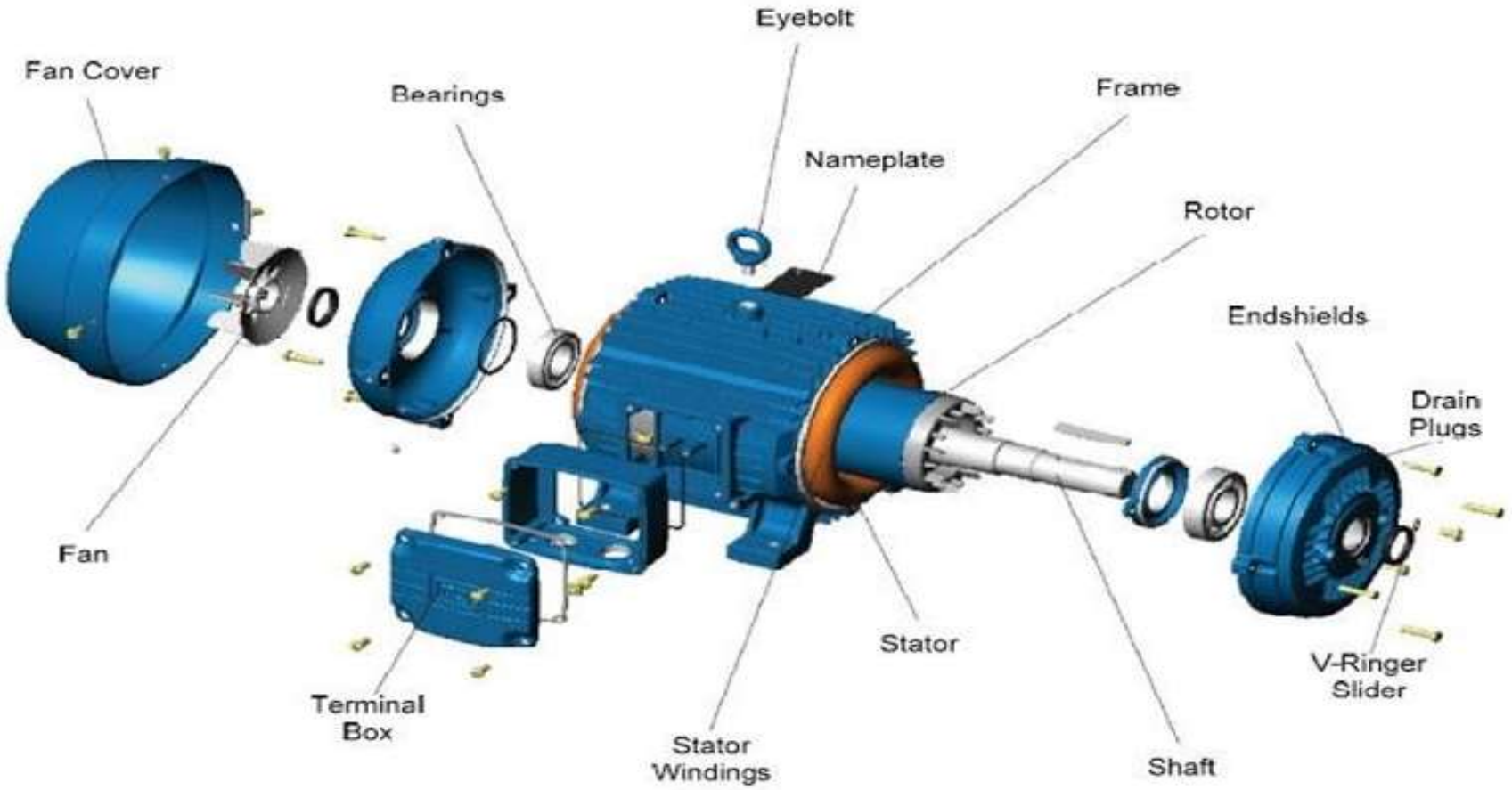
➢ **Stator:**

- It is stationary part of induction motor. It consists stator winding
- It is housed on the motor frame.

➢Rotor:-

- It is the rotating part of the induction motor.
- It is housed on the shaft of the induction motor.
- It has two ends, one is called **Driving end** and another is called **non Driving end**.
- ***Mechanical load*** is connected on driving end while ***cooling fan*** is connected on non-driving end.
- Both the ends are connected with bearings for free rotation means of reduced friction losses.

Assembly of 3-Phase Induction Motor



Three Phase winding in Induction Machine

- In a three phase alternators, ***three different sets of windings*** are arranged in the armature (stator) slots in such a way that there exists a ***phase difference of*** $\mathbf{120°}$ between the induced e.m.f in them.
- Each set of windings is for each phase of three phase induced e.m.f.
- Thus the three phase induced e.m.f has three different e.m.f given by,

$$e_R = E_m \sin(\omega t)$$

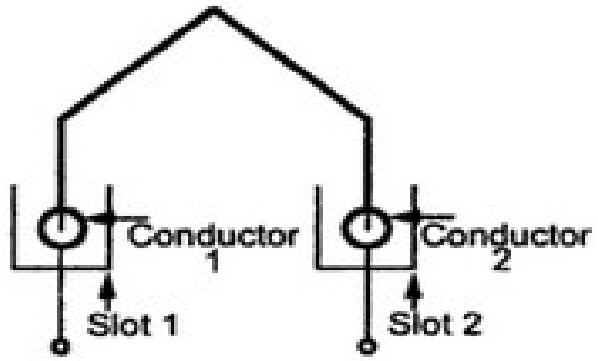
$$e_Y = E_m \sin(\omega t - 120°)$$

$$e_B = E_m \sin(\omega t - 240°)$$

AC Winding Terminologies

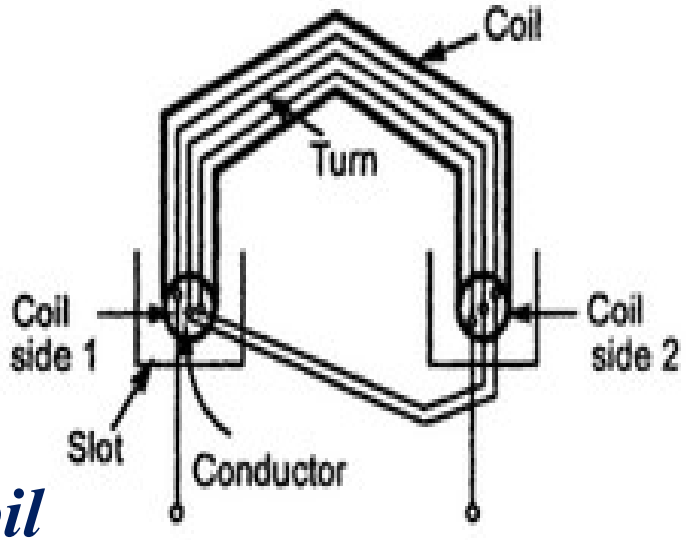
1. **Conductor:-** The part of the wire, which is under the influence of the magnetic field and responsible for the induced e.m.f. is called active length of the conductor. The conductors are placed in the stator slots.
2. **Turn :-** A conductor in one slot, when connected to a conductor in another slot forms a turn. So two conductors constitute a turn.
   - Two conductors connected in series and separated from each other by a pole pitch so that the emf induced will be additive.

Turn



3. Coil :- As there are number of turns, for simplicity the **number of turns are grouped together** to form a coil. Such a coil is called ***multi-turn coil.*** A coil may consist of single turn called ***single turn coil***.

4. Coil Side :- Coil consists of many turns. Part of the coil in each slot is called coil side of a coil.



Multi-turn coil

5. Pole Pitch:- It is center to center ***distance*** between the two adjacent poles.

- For one rotation of the conductors, **2 poles** are responsible for **360° electrical** of e.m.f, **4 poles** are responsible for **720° electrical** of e.m.f and so on.
- **1 pole** is responsible for **180° electrical** of induced e.m.f so 180° electrical is also called ***one pole pitch***.
- Practically how many slots are under one pole which are responsible for 180° electrical, are measured to specify the pole pitch.

Example:- Consider **2 pole**, **18 slots** armature of an alternator. Then under **1 pole** there are **18/2** i.e. **9 slots**. pole pitch is 9 slots or 180° electrical. means 9 slots are responsible to produce a phase difference of 180° between the e.m.f.s induced in different conductors.

- ***Number of slots/pole*** is denoted as **'n'**

Pole pitch	**=**	**180° electrical**
	=	**Slots per pole (no. of slots/P) = n**

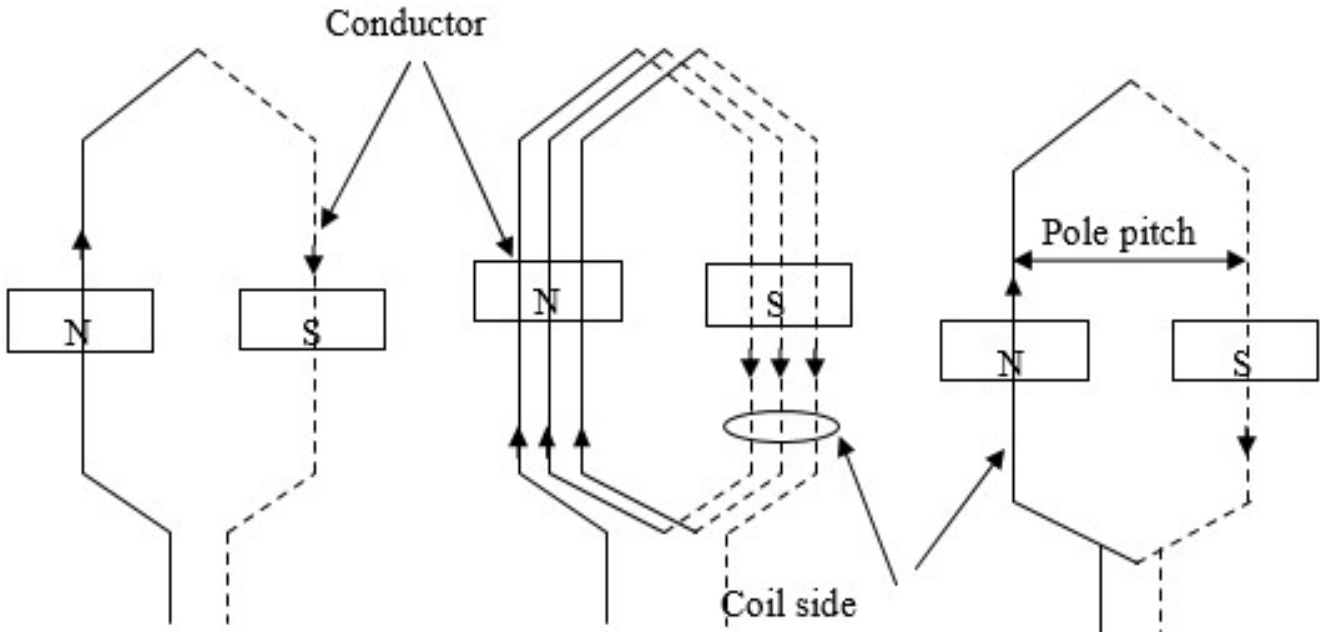
6. Slot angle (β) :- The ***phase difference*** contributed by ***one slot*** in degrees electrical is called **slot angle β**.

- As slots per pole contributes 180° electrical which is denoted as 'n',
- we can write, 1 slot angle(β) = 180°/n

In the above example,

$$n = \frac{18}{2} = 9, \quad \text{while} \quad \beta = \frac{180^\circ}{n} = 20^\circ$$

- **Note :** This means that if we consider an induced e.m.f in the conductors which are placed in the slots which are adjacent to each other, there will exist a phase difference of β° in between them.
- While if e.m.f. induced in the conductors which are placed in slots which are 'n' slots distance away, there will exist a phase difference of 180° in between them.



7. Coil span:- Coil span is the distance between two coil sides measured in terms of slots.

- Coil span = winding pitch/slot angle;
- For full pitched winding 180°/slot angle.

AC Windings

➢Generally open windings except when delta connected.

- **Classification:**
- ***Based on:***
  - **Supply:** 1- Φ & 3 - Φ
  - ***Placing:*** Concentrated and Distributed
  - ***No. of layers:*** Single and Double layer
  - ***End connection:*** Lap, wave and concentric
  - ***The pitch:*** Full pitched and short pitched

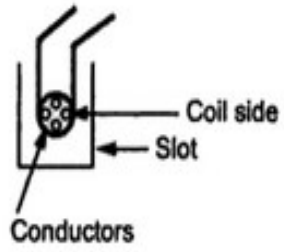
Types of Three Phase Winding

❖In general, the types of three phase winding used in a-c machines are,

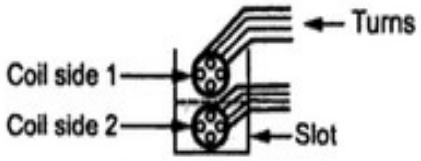
1. ***Single layer and double layer***
2. ***Full pitch and short pitch***
3. ***Concentrated and distributed.***

✓Single layer and double layer

- If ***a slot*** consists of ***only one coil side***, winding is said to be **single layer**.
- While there are ***two coil sides per slot***, one at the bottom and one at the top the winding is called **double layer**.



(a) Single layer



(b) Double layer

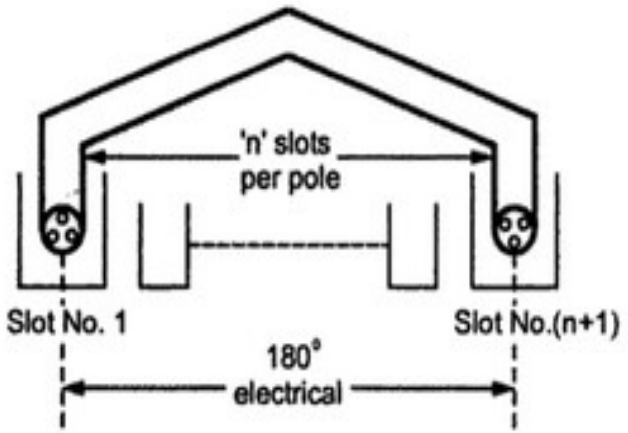
➢***A lot of space gets wasted in single layer hence in practice generally double layer winding is preferred.***

✓Full pitch and Short pitch

✓ Full Pitch

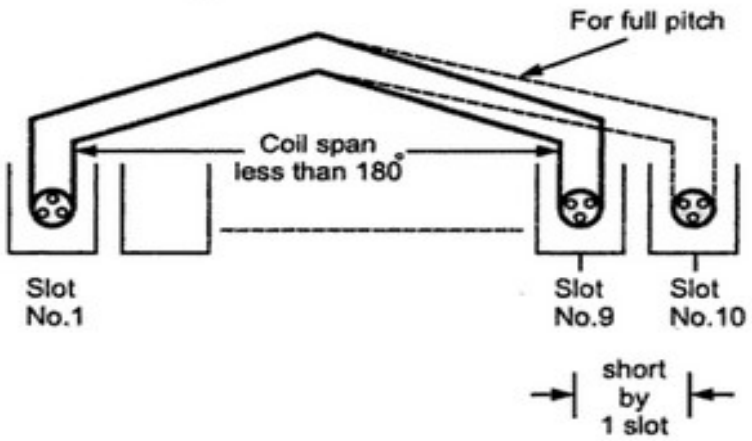
- If ***coil side in one slot*** is **connected** to ***a coil side in another slot*** which is ***one pole pitch distance*** away from first slot, the winding is said to be **full pitch winding** and **coil** is called **full pitch coil**.
- ***Example:-*** In 2 pole, 18 slots alternator, the pole pitch is n = 18/2 = 9 slots.
- So if coil side in slot No. 1 is connected to coil side in slot No. 10 such that two slots No. 1 and No. 10 are one pole pitch or n slots or 180° electrical apart, the coil is called ***full pitch coil***.

- **Coil Span:-** It is distance on the periphery of the armature between coil sides of a coil.
- It is usually expressed in terms of ***number of slots or degrees electrical.***
- So if coil span is 'n' slots or 180° electrical the coil is called full pitch coil.



✓Short Pitch

- If coils are used in such a way that coil span is slightly ***less than*** a pole pitch i.e. less than 180° electrical, the coils are called, **short pitched coils** or **fractional pitched coils**.
- Generally coils are shorted by ***one or two slots***.
- So in 18 slots, 2 pole alternator instead of connecting a coil side in slot No. 1 to slot No.10, it is connected to a coil side in slot No. 9 or slot No. 8, coil is said to be short pitched coil and winding is called short pitch winding.



- **Advantages of Short Pitch Coils (Chording)**

In actual practice, short pitch coils are used as it has following advantages

a) **The length required for the end connections of coils is less i.e. inactive length of winding is less. So less copper is required. Hence economical.**

b) **Short pitching *eliminates* high frequency harmonics which distort the sinusoidal nature of e.m.f, Hence waveform of an induced e.m.f is more sinusoidal due to short pitching.**

c) **As high frequency harmonics get eliminated, eddy current and hysteresis losses which depend on frequency also get minimized. This increases the efficiency.**

✓Concentrated and distributed Winding

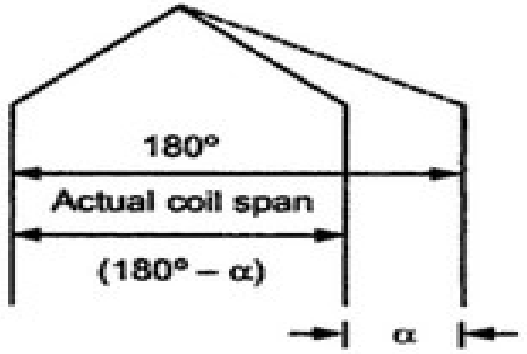
- In three phase alternators, there are three different sets of windings, each for a phase.
- So depending upon the total number of slots and number of poles, we have certain slots per phase available under each pole.
- This is denoted as **'m'**
  - **m =** ***Slots per pole per phase***
    = n/Number of phases
    = n /3 (generally no. of phases is 3)
- ***For example:-*** in 18 slots, 2 pole alternator we have;
  - n = 18/2 = 9
  - m = n/3 = 9/3 = 3
- So we have 3 slots per pole per phase available.

- Now let **'x'** number of conductors per phase are to be placed under one pole. And we have 3 slots per pole per phase available. But **if all 'x' conductors per phase** are placed **in one slot** keeping remaining ***2 slots per pole per phase empty*** the winding is called **concentrated winding.**
- So in concentrated winding all conductors or coils belonging to a phase are placed in one slot under every pole.
- But in practice, an attempt is always made to use all the slots per pole per phase available for distribution of the winding.
- So if **'x' conductors per phase** are distributed amongst the 3 slots per phase available under every pole, the winding is called **distributed winding**.
- So in distributed type of winding all the coils belonging to a phase are well distributed over the **'m'** slots per phase, under every pole.
- Distributed winding makes the waveform of the induced e.m.f more sinusoidal in nature. Also in concentrated winding due to large number of conductors per slot, heat dissipation is poor.
- ***Key Point:-*** So in practice, **double layer**, **short pitched** and **distributed type** of winding is preferred for the alternators.

Pitch factor or coil span factor (Kc)

- In practice short pitch coils are preferred. So coil is form by connecting one coil side to another which is less than one pole pitch away. So actual coil Span is less than 180°. The coil is generally shorted by one or two slots.
- ***Key point*** :- The angle by which coils are short pitched is called angle of short pitch denoted as **'α'.**
- α = angle by which coils are short pitched. As coils are shorted in terms of number of slots i.e. either by one slot, two slots and so on and slot angle is β, then angle of short pitch is always a multiple of the slot angle β.
  - **α = β x Number of slots by which coils are short pitched.**
  - **or 180° — Actual coil of the coils.**
- ***Key Point:-*** Now the factor by which, induced e.m.f. gets reduced due to short pitching is called pitch factor or coil span factor denoted by **Kc**.

$$Kc = Cos(\alpha/2)$$



Distribution Factor (Kd)

- Similar to full pitch coils, concentrated winding is also rare in practice.
- Attempt is made to use all the slots available under a pole for the winding which makes the nature of the induced e.m.f. more sinusoidal. Such a winding is called distributed winding.
- ***Key Point:-*** The factor by which there is a reduction in the e.m.f. due to distribution of coils is called distribution factor denoted as **'Kd'.**

$$\therefore \quad K_d = \frac{\sin\left(\frac{m\beta}{2}\right)}{m\sin\left(\frac{\beta}{2}\right)}$$

Where

m = Slots per pole per phase

β = Slot angle $= \frac{180^\circ}{n}$

n = Slots per pole

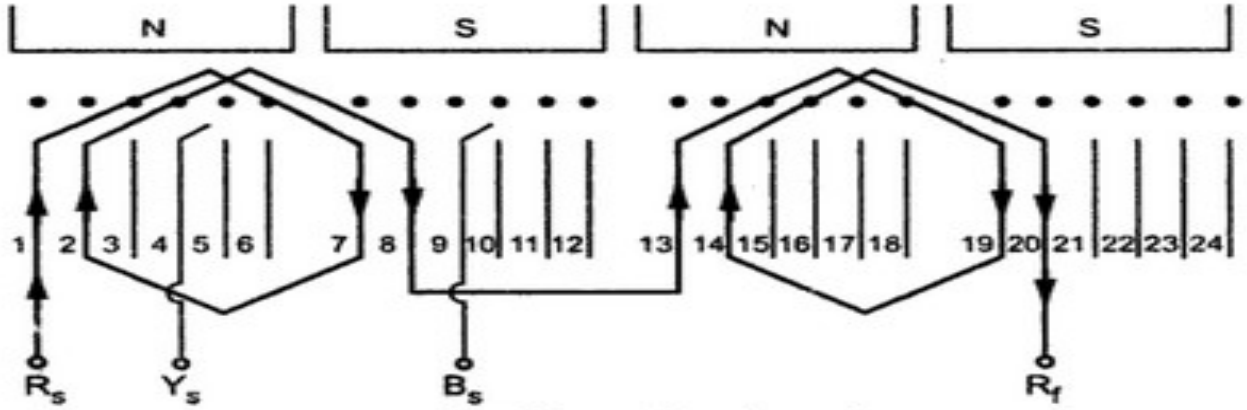
- ***Example 1:*** ***Draw the developed/winding diagram*** for **full pitch** stator winding of a **three phase**, **4 pole**, **24 slots** alternator. Assume single layer winding, Lap winding and of distributed type.
  - *Note:* This example will explain all the winding terminologies discussed earlier.

➢**Solution:** p = 4, 24 slots, 3-phase

- **n = slots per pole = 24/4 = 6**
- **m = slots per pole per phase = 6/3 = 2**
- **β = slot angle = 180°/n = 180°/6 = 30°**

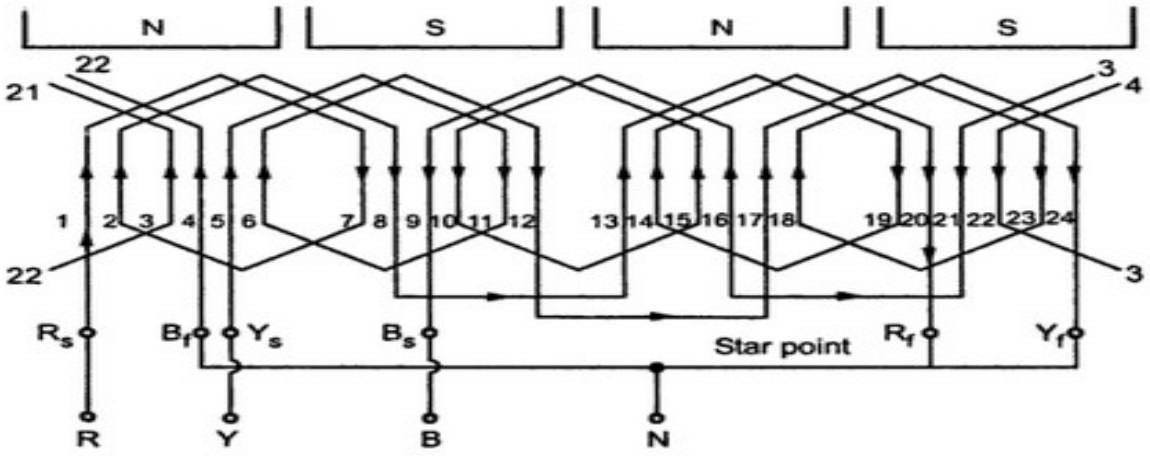
- ***Full pitch coils are to be used so if phase 1 say, R is started in slot 1, it is to be connected to a coil in slot 7 so that coil span will be 6 slots i.e. 'n' slots i.e. 1 pole pitch.***

- As distributed winding is to be used, both the slots per pole per phase (**m = 2**) available are to be used to place the coils.
- And all coils for **one phase** are to be connected **in series**. So from **slot No.7** we have to connect it to coil **slot No.2** and slot No.2 second end to slot **No.8** and so on.
- After finishing all slots per phase available under first pair of pole, we will connect coil to **Slot No.13 under next pole** and winding will be repeated in similar fashion. starting end **Rs** and final end **Rf** winding for **R-phase** are taken **out finally**. Connections for R-phase only are shown below.



All coils per phase in series

- Now we want to have a phase difference of **120°** between **'R'** and **'Y.** Each slot contributes 30° as β = 30°.
- So **start of 'Y' phase** should be **120°** apart from **start of 'R'** i.e. **4 slots** away from **start of R**. So start of 'Y' will be in **slot 5** and will get connected to **slot No.11** to have full pitch coil.
- Similarly **start of 'B'** will be further **120°** apart from **'Y'** i.e. **4 slots** apart start of 'Y' i.e. will be in **slot No.9** and will continue similar to 'R.
- Finally all ***six terminals of three sets will be brought out*** which are connected either in **star or delta** to get **three ends R, Y and B outside** to get **three phase supply**. The entire winding diagram with star connected windings is shown as;



➢ ***Example 2:*** *Draw the developed winding diagram of a 3 phase induction motor with 18 slots, 2 poles, single layer, full pitched winding with delta connection.*

Soln:- **Pole pitch (n) = no. of conductor / pole = 18/2 = 9**

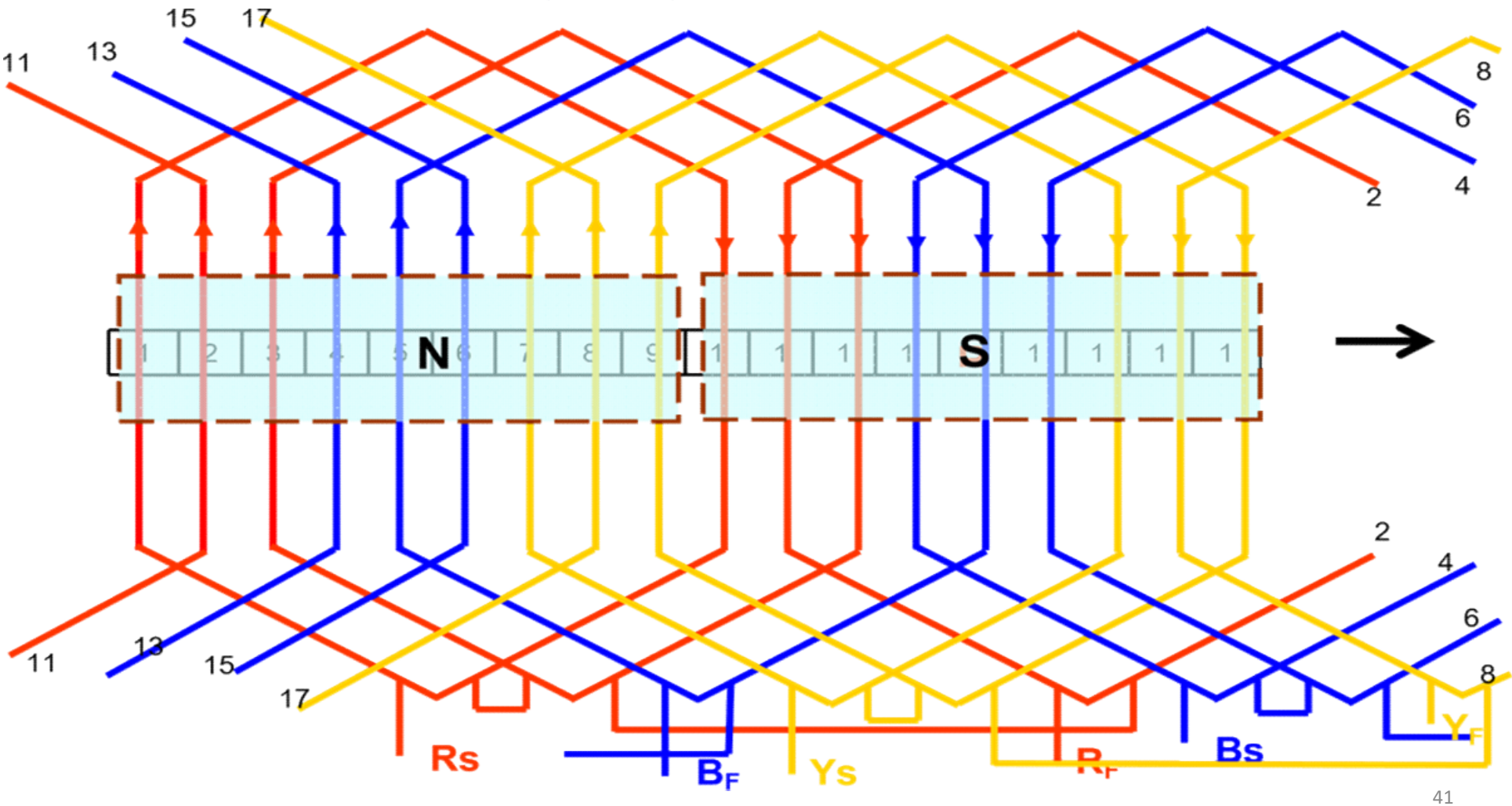
- **No. of slots per pole per phase (m) = 18/(2 x 3) = 3**
- **Slot angle (β) = 180/ pole pitch = 180°/9 = 200**
- **Full pitched winding = coil span = 180°**
- **Coil span = winding pitch/slot angle = 180/20 = 9 slots**

Winding Table

Phase	1st pole	2nd pole
R	1 + 9 = 10	11+ 9 = 20 (2)
	3 + 9 = 12	
Y	7 + 9 = 16	17 + 9 = 26 (8)
	9 + 9 = 18	
B	13 + 9 = 22(4)	
	5 + 9 = 14	15 + 9 = 24 (6)

➢ **Connections: Rs = 1, Ys = 1 + 120/slot angle = 1+120/20 = 7; Bs = 1 + 240/slot angle = 1+ 240/20 = 13**

Winding Diagram



Windings Connection of Induction Motor

➢Stator Windings:

- Star connected
- Delta connected

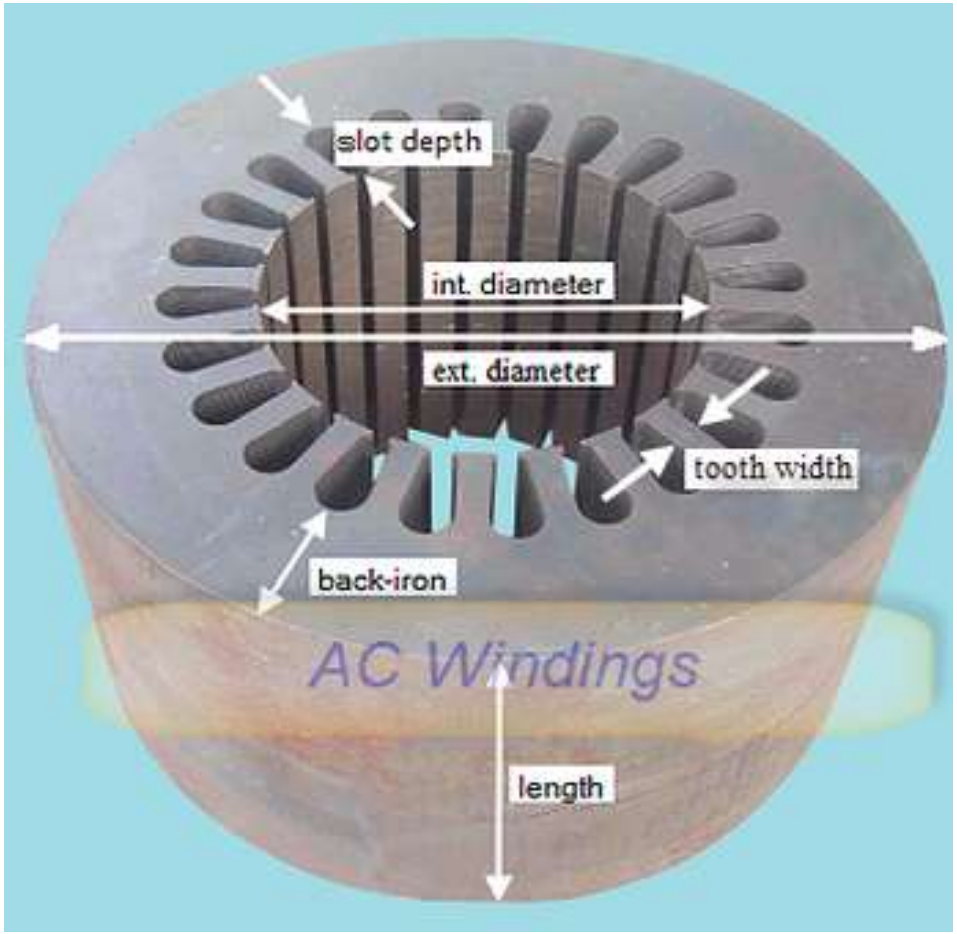
➢Rotor Windings:

- It is wound as rotor bars and short circuited at both the ends through end rings.



What data you need?

- Line to line voltage (V),
- Frequency (Hz),
- Star or delta connection,
- Number of poles or rpm,
- All other nameplate data if available,
- Stator slots,
- Core internal diameter (mm),
- Core external diameter (mm),
- Core length,
- Number and width of air vents if exist,
- Back-iron (mm),
- Tooth-width (mm)



➢Materials needed for Motor winding

- Copper wire
- Isolate paper (for insulation)
- Stator lacing thread (for tying windings)
- ***Spray Lubricant***
- ***Motor lacquer (for smoothing)***

Thank you for your Attention!!

